

OHIDA BINTE AMIN

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EDUCATION

Northeastern University, Boston, MA, USA

September 2022 - Present

PhD in Computer Science / Personal Health Informatics

Northeastern University, Boston, MA, USA

April 2026

MS in Computer Science (not conferred)

Khulna University of Engineering & Technology, Bangladesh

March 2020

BS in Computer Science & Engineering

EXPERIENCE

Khoury College of Computer Sciences, Northeastern University

September 2022 - Present

Graduate Research Assistant (PhD Candidate) at Cognitive Systems Lab

- Developing a multimodal stress modeling framework using physiological, audio, and video data; implemented a Multi-Layer Perceptron to learn shared latent representations and explored probabilistic graphical modeling to capture dependencies between affective states, integrating ordinal regression with pairwise comparison learning (Bradley–Terry) to improve robustness to noisy self-reports
- Deriving sleep phenotypes from wearable and under-mattress sensor data using a Hidden Markov Model, identifying fragmented vs. consolidated sleep states based on sleep patterns, and analyzed their physiological associations to characterize patterns linked to delirium risk
- Building interpretable stress prediction models by integrating behavioral rules from passive sensing data with physiological biomarkers, derived from an Support Vector Model (SVM) from lab stressors with an AdaBoost model across temporal windows, achieving performance with ROC-AUC of 78.75%
- Conducted a controlled digital phenotyping stress study across research and consumer-grade wearables, applying SVM with RBF kernel and Random Forest where Garmin Forerunner 55s achieved highest performance, outperforming Empatica E4 by 6.2%
- Analyzed stress–sleep relationships using ANOVA and linear mixed effects model revealing greater multiday sleep efficiency variability among high-stress individuals, addressing gaps in capturing different social determinants in sleep analysis.

Samsung Research America

May 2024 – August 2024

Data Science & Machine Learning Intern at Samsung Design Innovation Center

- Designed methods to aggregate and analyze unstructured response data for dyslexia support, using TextBlob for sentiment analysis and Term Frequency–Inverse Document Frequency (TF–IDF) vectorization for text feature extraction.
- Built a feed-forward neural network MLP model to predict personalized support needed for Dyslexic individuals with smartphones for developing a prototype with Streamlit framework.

Baylor College of Medicine, Rice University

April 2021 - December 2021

Research Fellow at The Fatima Al-Fihri Predoctoral Fellowship

- Worked on automatic annotation of 3D genomic datasets using self-supervised deep learning approaches & conducted comparative study with different clustering techniques including K-means and DBSCAN on HiC features.

Robi Axiata Limited, Dhaka, Bangladesh

July 2020 - February 2022

Data Science & Machine Learning Engineer

- Developed a complaint-classifier ML model using a feed-forward neural network MLP with ReLU and Soft-max layers, handling training, validation, and deployment to resolve customer complaints with zero human intervention.
- Built linear regression-based predictive model for suggesting suitable offers to the customers (like recharge, data pack, etc.) development, training, & validation.

Abelling, Dhaka, Bangladesh
Machine Learning Intern

June 2020 - July 2020

- Creating documentation for ML products and reviewing AI-assisted data labeling APIs & Abelling client data analysis by applying statistical analysis.

RECENT PUBLICATIONS & PRESENTATIONS

[Google Scholar Profile](#)

- A. Sathyanarayana*, **O.B. Amin***, J. An, and J.-P. Onnela, “*Examining the use of consumer wearable devices and digital tools for stress measurement in college students: a scoping review of methods*,” has been accepted for publication in JMIR mHealth and uHealth
- **O.B. Amin**, T.M. Tapera, R. Volpe, V. Mishra, and A. Sathyanarayana “*Extending Stress Detection Reproducibility to Consumer Wearable Sensors*,” in 2025 47th International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) **IEEE**, 2025. [[Paper Link](#)]
- **O.B. Amin**, V. Mishra, and A. Sathyanarayana, “*Investigating Social Interaction Patterns with Depression Severity across Different Personality Traits Using Digital Phenotyping*,” in 2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW). **IEEE**, 2023. [[Paper Link](#)]
- **O.B. Amin**, V. Mishra, and A. Sathyanarayana, “*The Impact of Stress and Sleep: Capturing Multi-day Patterns*,” in 2024 38th Annual Meeting of SLEEP, vol. 47, issue supplement1, 2024. [[Abstract Link](#)]

SKILLS

Language	Python, C, C++, R, Shell, Swift, SQL, Java
Frameworks/Tools	TensorFlow, PyTorch, Keras, Scikit-learn, PySpark, Pandas, NumPy, Matplotlib Oracle DBMS, MySQL, PyMongo, StatsModels, DEAP, Streamlit, Flask, Odoo, SkFuzzy, MATLAB Fuzzy Toolbox, KML, KML3D (R), OpenAI API
Domain	Machine Learning, Deep Learning, Digital Phenotyping, Digital Biomarkers, Mobile Sensing, Signal Processing, Data Science, Natural Language Processing, Human Computer Interaction
Wearable/Medical Devices	Empatica E4, Garmin, Polar Chest Straps, Biopac MP160 System

AWARDS & ACHIEVEMENTS

- **SRS Diversity Membership Award 2023** – Sleep Research Society
- **Start-Up Fund Recipient** – Northeastern University, Khoury College of Computer Sciences
- **Selected Research Fellow** – The Fatima Al-Fihri Predoctoral Fellowship 2021

RELEVANT COURSES

Programming Design and Paradigm, Algorithms, Personal Health Interface Design and Development, Information Visualization Theory and Application, Mobile Application Development, Empirical Research Methods for Human Computer Interaction, AI for Human Computer Interaction, Understanding Users, Evaluating Health Technologies, Biostatistics in Public Health

TEACHING

CS 5200: Database Management Systems , Northeastern University	Summer 2025 – Spring 2026
<i>Lead Teaching Assistant (with Prof. Martin Schedlbauer)</i>	
PHTH 2210: Foundations of Biostatistics , Northeastern University	Summer 2026
<i>Teaching Assistant (with Prof. Robert Lindsey)</i>	